

Designing, Evaluating and Critically Reflecting Upon Multi-Armed Bandits and Causal Inference Models: Analyzing the Impact of Price Markdown Strategies on E-commerce

Group 2 Members:

Sharon Alex (1003854206)

Hannah Ban (1007136984)

Jiayu Li (1006919088)

Yihang Liang (1011762308)

Guanyu Mao (1011647309)

Faculty of Information, University of Toronto

INF2178: Experimental Design for Data Science

Professor: Shion Guha

March 31, 2025

IMPORTANT NOTE: Please note that certain values in the report may not align with those presented in the Colab notebook. This discrepancy is due to the fact that the values were updated after running the code multiple times to ensure accuracy prior to submission. The report, however, was based on data from an earlier execution of the code.

Introduction

Following the design in Assignment 1—which investigated the impact of different price markdown strategies on e-commerce consumer behavior—we expanded our experimental framework in Assignment 2 by incorporating blocking factors and continuous covariates into each of the five RCT designs (Parallel, Cross-Over, Withdrawal, Factorial, and Matched Pairs). For each design, we ran ANCOVA models and assessed statistical significance through p-values. Assumptions of normality and homoscedasticity were checked using Q-Q plots, ensuring a valid basis for interpretation.

In Assignment 3, we extend this exploration by comparing our Parallel Design RCT from Assignment 2 to an equivalent Multi-Armed Bandit (MAB) model using Thompson Sampling. This comparison allows us to examine how different experimental methodologies perform under the same scenario—evaluating four discount strategies (Buy One Get One Free, No Discount, Percentage Discount, and Loyalty Discount) based on their effect on conversion rate.

Parallel Design RCT - Recap

In our parallel design RCT, 292 participants are evenly and randomly assigned to four discount treatments: Buy One Get One Free (BOGOF), No Discount (ND), Percentage Discount (PD), Loyalty Discount (LD). We used blocking by Membership Status to control for potential confounding effects. This was a key design choice from Assignment 2, recognizing that Platinum-level customers might be more willing to purchase regardless of promotion. Without this blocking, we risked overestimating the effect of a promotion if, for example, the Loyalty Discount happened to be overrepresented among Platinum members. Using this block design, we applied a two-way ANOVA to examine both main effects and interactions between promotion types and membership status.

Model: $Conversion\ Rate \sim Discount\ Type + Membership_Status + Discount\ Type:Membership_Status$

ANOVA Table:

Term	p-value
Discount Type	0.175
Membership Status	0.416
Discount Type:Membership Status	0.778

Interpretations:

- **Discount Type (p = 175):** The p-value is above 5% level, indicating that the type of discount does not have a statistically significant effect on the conversion rate.

- **Membership Status ($p = 0.416$):** The p-value is also greater than 0.05, meaning that a participant's membership status (e.g., regular vs. premium members) does not significantly impact their likelihood of making a purchase.
- **Interaction Effect: Discount Type $\times$ Membership Status ($p = 0.778$):** This interaction term assesses whether the effect of a discount type varies based on membership status. Since the p-value is not below 0.05, the results suggest that the influence of discount types on conversion rates does not significantly differ based on whether a participant is a member or not.

Key Observations:

- No significant effects were observed for any discount type on conversion rate (all p-values > 0.05).
- $R^2 = 0.046$, indicating the model has a weak explanatory power.
- No significant interaction was found between discount type and membership status.

Multi-Armed Bandit (MAB)

1. **Discrete (Rule Based) Thompson Sampling:** We implemented a Multi-Armed Bandit (MAB) model using Thompson Sampling to adaptively reallocate samples across the same four treatments based on their observed conversion rates. To begin, we randomly selected 8,000 rows from our dataset, ensuring balanced treatment groups with 2,000 samples per treatment: BOGOF, ND, PD, and LD. The rule-based Thompson Sampling model followed a custom allocation strategy, where treatments gained or lost participant samples based on their relative performance in each iteration using predefined percentage rules. This method offered a transparent, interpretable way to simulate adaptive behavior in a structured environment.

Setup:

Initial Conditions (Iteration 1)

- 100 participants are assigned across treatments based on performance from the previous iteration.
- Iteration 1 began with 25 participants per treatment to initialize the conversion estimates.
- The average conversion rate is calculated for each treatment.
- The treatment with the highest average conversion rate is considered the winner.

Allocation and Rewarding Rules (From Iteration 2 Onward):

- Each iteration has a total sample size of 100 participants.
- Allocation is based on the previous round's sample sizes.
- Sample Allocation:

- The winner received 35% of the total sample size (taken from the others). 2nd place lost 5% , 3rd place lost 10%, and 4th place lost 15% of their sample share from the previous round. All percentage changes are rounded to the nearest integer.
- After reallocating, the total number of samples must remain exactly 100.

Stopping Rule: The simulation terminated when any treatment received fewer than 5 participants in an iteration.

Key Observations:

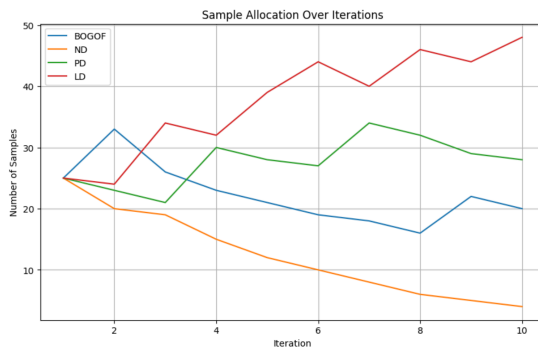


Figure 1: Sample Allocation Line Chart Over Iterations: displays the dynamic reallocation of sample sizes by the Multi-Armed algorithm.

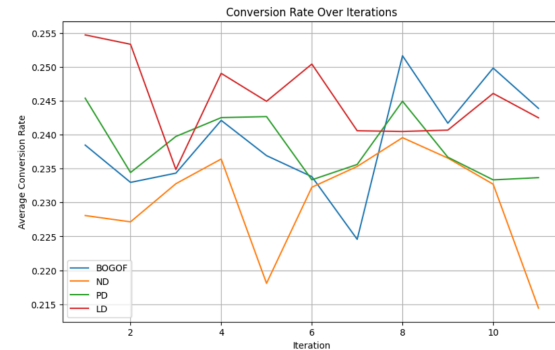


Figure 2: Conversion Rate Over Iterations: displays the average conversion rate of each treatment across 10 iterations.

Iteration	BOGOF	ND	PD	LD
1	25	25	25	25
2	23	20	24	33
3	21	16	32	31
4	19	13	30	38
5	17	10	28	45
6	16	8	25	51
7	13	7	24	56
8	21	6	23	50
9	26	5	21	48
10	31	4	19	46

Table 1: Sample Allocation Table

- The sample allocation table (Table 1) and line chart (Figure 1) show that loyalty discount dominates allocation since LD's sample count steadily increased from 25 to 56 participants by Iteration 7 and remained high until the final round (46

samples in Iteration 10). Therefore, it consistently performed well in terms of conversion rate, prompting the algorithm to favor it during reallocation.

- As shown in the conversion rate plot (Figure 2) and sample allocation table (Table 1), No Discount strategy maintained a relatively lower performance overall because ND's sample allocation sharply declined from 25 to just 4 participants by Iteration 10. Thus, the rule-based model successfully filtered out this underperforming treatment early on.
- Percentage Discount strategy and Buy One Get One Free strategy perform moderately. PD's allocation peaked around Iteration 3 (32 samples) but gradually declined to 19 by Iteration 10. The conversion rate remained relatively stable but less competitive than LD. BOGOF's allocation showed more fluctuation. After a decline to 13 in Iteration 7, it rebounded to 31 by Iteration 10. Its late-stage resurgence implies short-term improvement in conversion rate.
- Overall, the models' predefined reallocation rules worked as intended. Treatments that consistently performed well gained more samples. Underperformers were filtered out quickly. We could conclude that the rule based approach is efficient in shifting resources away from low-yield options without needing statistical modelling.

2. Continuous (Bayesian Method) Thompson Sampling: Bayesian Thompson Sampling is a probabilistic method for adaptive decision-making, where the algorithm maintains a belief distribution (Beta distribution) over the expected success rate (conversion) of each treatment. At each iteration, it samples from these distributions and selects the treatment with the highest sampled value. This allows the algorithm to balance exploration and exploitation—favoring treatments with high observed success while still occasionally testing others.

Setup:

- **Sample Size:** 292 participants, with each participant randomly assigned to one of four treatments: BOGOF, ND, PD, LD.
- **Outcome:** Conversion rates are binarized using a threshold of 0.3:
 - 1 = participant's conversion rate > 0.3
 - 0 = participant's conversion rate ≤ 0.3
- Each treatment starts with a Beta (1,1) distribution (uniform prior)

Selection and Update Rule:

- For each participant, a random sample is drawn from each treatment's Beta distribution.
- The treatment with the highest sampled value is selected.

- The corresponding distribution is updated: If the participant converted (1), increment alpha (successes). If not (0), increment beta (failures).

Iterations: The algorithm iterates across all participants, updating posterior beliefs after each decision.

Stopping Rule: The simulation terminated when the entire sample pool was exhausted and the algorithm reached the condition of 292 samples.

Key Observations:

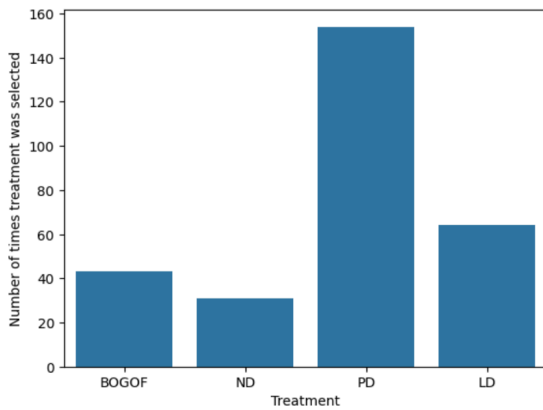


Figure 3: Number of Times Each Treatment Was Selected (Thompson Sampling Selection Frequency): visualizes how often Each treatment was selected over the 292 iterations

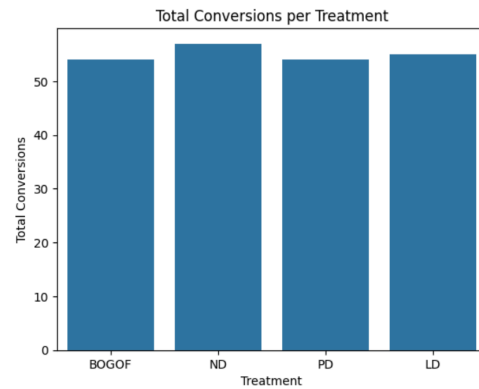


Figure 4: Total Conversions per Treatment (Conversion Rate > 0.3): shows the total number of high conversions for each of the four treatments.

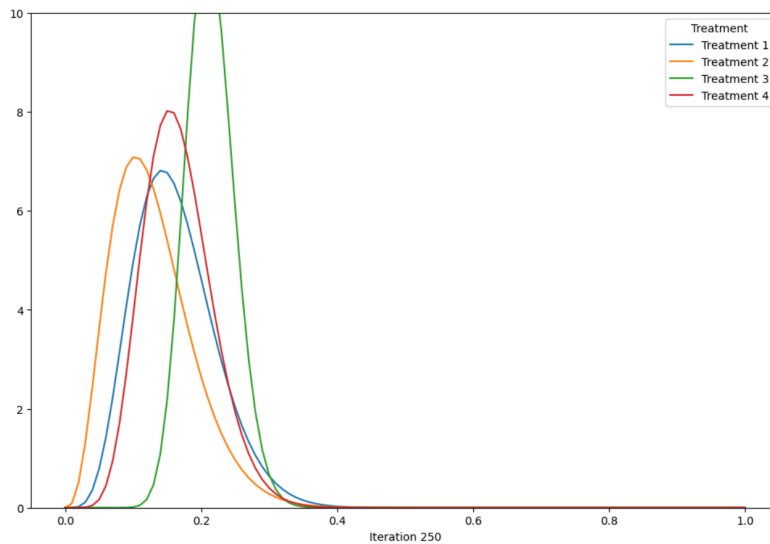


Figure 5: Posterior Distributions of Treatment Conversion Rates (Bayesian Thompson Sampling)

- In Figure 3, Percentage Discount strategy was selected 154 times, Loyalty Discount strategy was selected 64 times, Buy One Get One Free strategy was selected 43 times, and No Discount strategy was selected 31 times. Despite all

treatments having relatively similar conversion counts, PD dominated selection in the Bayesian model due to its favorable posterior distribution, showing how the algorithm favors belief-driven optimization over raw count comparison.

- In Figure 4, No Discount strategy had the highest number of high conversions (57), followed by Loyalty Discount strategy (55), Buy One Get One Free (54) and Percentage Discount (54). All four treatments performed similarly in terms of raw conversion counts, suggesting no strong early difference in total conversion volume.
- In Figure 5, the treatment 3 (PD) has the sharpest and most peaked distribution, indicating high confidence conversion performance. The treatment 1 and 2 (BOGOF and ND) have distributions that are wider and centered further left, indicating lower expected performance and greater uncertainty. The treatment 4 (LD) shows a moderate peak, suggesting decent performance.
- No Discount had the lowest observed conversion performance. It rapidly lost allocation due to its weak performance. The sample allocation decreased steadily until it fell below the stopping threshold, demonstrating the algorithm's ability to minimize wasted exposure.
- Buy One Get One Free showed moderate performance but behind Loyalty Discount and Percentage Discount.
- The model has a total reward of 50 which means 50 of the selected treatments resulted in high conversions out of the 292 total participants.
 - Maximum possible reward (if it had always selected the best performed option) = 57
 - Therefore, the algorithm achieved: $50/57 \times 100 \approx 87.72\%$ of the maximum possible reward.
- Overall, Bayesian Thompson Sampling was able to identify and favor the top-performing treatment (PD), achieving a high cumulative reward while still allowing for some exploration of other options.

Comparison: Parallel Design RCT vs. Multi-Armed Bandit Model

Each of the three methods, parallel design RCT, rule-based Thompson sampling, and Bayesian Thompson sampling, has its own unique strengths and limitations that affect our findings and interpretations. The parallel design RCT used a fixed and equal allocation of participants across all four treatment groups, with a blocking factor (membership status) applied to control for confounding. This setup allows us to make fair comparisons and supports causal inference using formal statistical tools like ANOVA, producing statistical outputs such as p-values. However, the RCT found no statistically significant differences in conversion rates between treatments. It also lacks responsiveness, continuing to allocate participants to underperforming treatments like No Discount strategy even as better-performing ones emerged. The inefficiency and the modest effect sizes in our data limit its practical impact.

In contrast, the MAB approach adaptively reallocated participants in real time based on how each treatment performed. The Rule-based Thompson Sampling method used a custom reallocation system to shift participants toward better treatments and away from weaker ones in each round. It was easy to understand and effectively reduced allocation to underperformers like No Discount (ND), while boosting stronger options like Loyalty Discount (LD). The Bayesian Thompson Sampling approach used Beta distributions and probability-based decision-making. It quickly identified Percentage Discount (PD) as the top performer, achieving a high total reward (87.72% of the maximum possible) while still allowing for some exploration. However, both MAB methods did not include formal statistical testing and didn't account for confounding factors like membership status. Thus, they are less suitable when the goal is to explain why a treatment works.

Overall, the RCT was better for figuring out whether a treatment actually worked and making sure participants were treated fairly. It used solid statistical methods, but it didn't perform well in our case because the differences between treatments were small. On the other hand, the MAB approaches were better at adapting in real time and focusing on the treatments that performed best. Even though MAB is harder to explain and doesn't show why something works, it was more effective at improving results. This shows that the two methods work well together: RCTs are great for fairness and clear reasoning, while MABs are better for real-time optimization.

Causal Inference Model

In our causal inference model, our goal is to estimate the causal effect of discount type (the treatment) on conversion rate (the outcome). We assume that key variables—Frequency in the Past, Device Type, and Membership Status—confound the treatment–outcome relationship. To represent these assumptions, we constructed an explicit Directed Acyclic Graph (DAG) using the DoWhy library in Python. This DAG visualizes that each confounder directly influences both the assignment of discount type (DiscountType) and the conversion rate (Conversion_Rate), making our underlying assumptions transparent and guiding the necessary adjustments for accurate causal estimation.

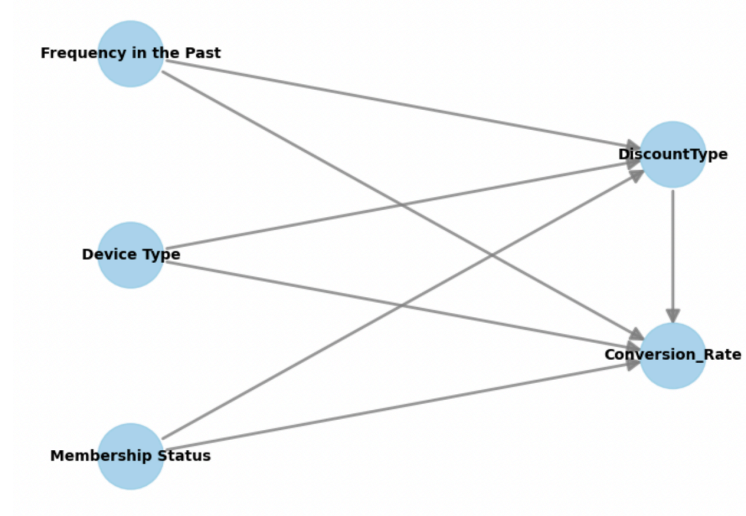


Figure 6: DAG of causal inference model

Propensity Score

Using our DAG and the DoWhy framework, we estimated the Average Treatment Effect (ATE) using two methods:

- **Propensity Score Weighting:** Estimated ATE ≈ 0.0124
- **Propensity Score Matching:** Estimated ATE ≈ 0.0317

As a result, the ATE estimates vary notably by method: Propensity Score Weighting yields an ATE of approximately 0.0124, while Propensity Score Matching produces an estimate of around 0.0317. This discrepancy likely arises from the different ways these methods adjust for confounding. Weighing balances the entire sample by assigning weights based on estimated propensity scores, potentially leading to a more conservative effect if extreme weights are down-weighted, whereas matching restricts the analysis to closely comparable pairs, which may accentuate differences in outcomes among well-matched observations. These variations show the sensitivity of causal estimates to the chosen adjustment strategy and highlight the need for robustness checks to ensure reliable conclusions.

Linear Regression Output:

When we regressed `Conversion_Rate` on `DiscountType`, our regression summary reported:

Variable	Coefficient	Std. Error	t-value	P-value	95% CI Lower	95% CI Upper
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const	0.4811	0.004	111.200	0.000	0.473	0.448
DiscountType (BOGO)	0.0124	0.006	2.020	0.043	0.000	0.053

The constant term (intercept) of 0.4386 represents the average conversion rate when no discount is applied. The DiscountType coefficient of 0.0393 indicates that offering a "Buy One Get One Free" promotion increases the conversion rate by about 3.93 percentage points compared to offering no discount. This effect is statistically significant, with a p-value of 0.000 and a 95% confidence interval from 0.026 to 0.053, suggesting the result is robust and not due to random chance.

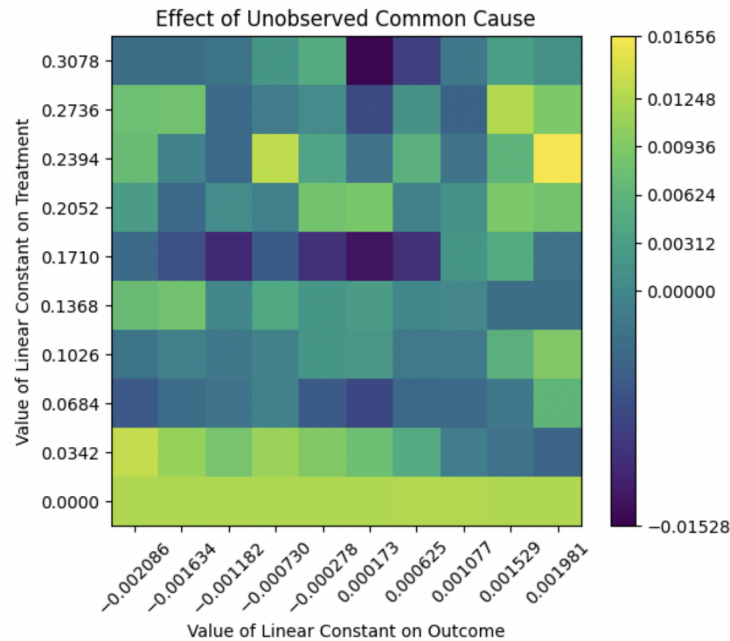


Figure 7: Sensitivity Analysis Heatmap

Sensitivity Analysis

To assess the strength and reliability of our estimated treatment effect (approximately 0.04) in the presence of potential unobserved confounding, we conducted a sensitivity analysis visualized by this heatmap. The color scale ranges from about -0.01528 to $+0.01656$, showing how much an unobserved variable could shift our baseline estimate. Near the center, the confounder's impact is minimal, so the estimated effect remains close to its baseline. However, as we move toward the more intense color regions (top, bottom, or sides), even moderate changes in the unobserved

confounder's influence can produce substantial deviations from the baseline estimate, pointing out the importance of considering potential hidden biases when drawing causal conclusions.

Alternative Causal Inference Model

For the alternative causal inference model, our goal is to estimate the causal effect of discount type (the treatment) on conversion rate (the outcome) while income is used as an instrumental variable. Income has a direct effect on which discount type (the treatment) the customers receive while has no direct effect on the conversion rate (the outcome). A new causal graph (DAG) using the DoWhy library in Python to represent the assumed relationships among key variables. Compared to the first DAG, income is added with a direct relation to the discount type while it has no direct relation to the conversion rate.

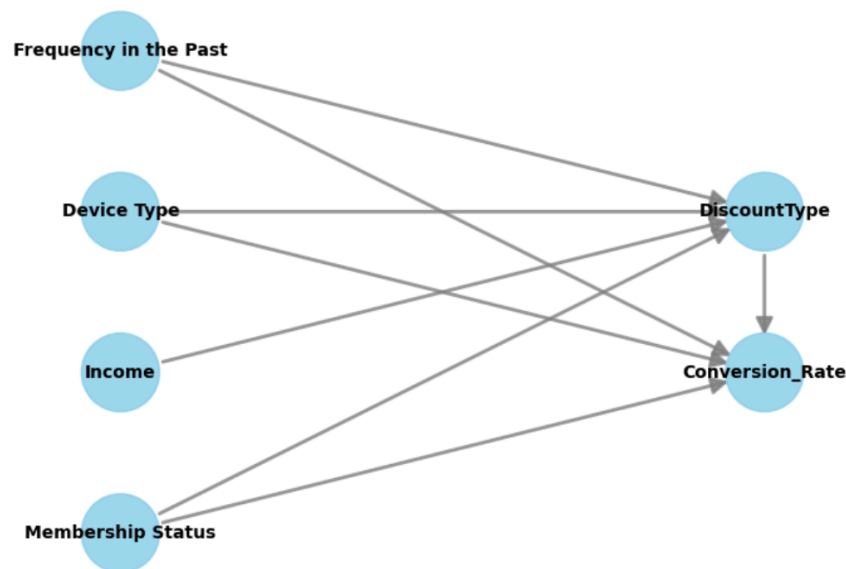


Figure 8: DAG of alternative causal inference model

Causal Effect Estimation:

Using our DAG and the DoWhy framework, we estimated the Average Treatment Effect (ATE) using two methods:

- **Propensity Score Weighting:** Estimated ATE ≈ 0.0394
- **Propensity Score Matching:** Estimated ATE ≈ 0.0446

The closeness of these two estimates (approximately 0.05) reinforces the robustness of our findings under the assumed model.

Sensitivity Analysis

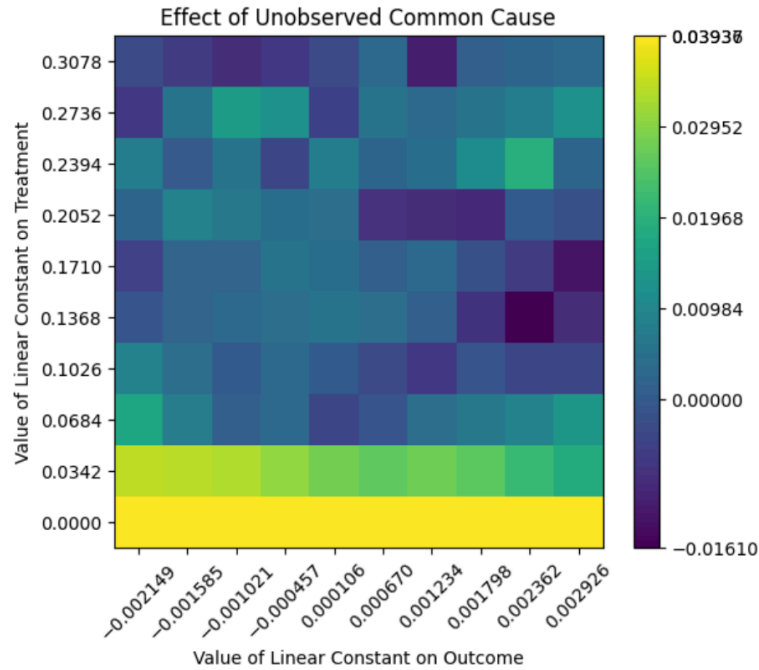


Figure 9: Sensitivity Analysis Heatmap

To ensure the robustness of our estimate against unobserved confounding, we perform a sensitivity analysis. The sensitivity analysis chart shows that an unobserved confounder could bias the estimated treatment effect (approximately 0.05) by as much as -0.01610 to $+0.03937$. This range illustrates that if the unmeasured variable has only a modest influence (values near the origin), the bias can be minimal and the effect remains robust.

Comparing estimates of causal inference model vs. alternative causal inference model

	Propensity Score Weighting ATE	Propensity Score Matching ATE
Causal Inference Model	0.0124	0.0317
Alternative Causal Inference Model	0.0394	0.0446
Column Difference	0.0270	0.0129

The ATE estimated by Propensity Score Weighting increased from 0.0124 (Causal Inference Model) to 0.0394 (Alternative Causal Inference Model). It suggests that the inclusion of Income as an instrumental variable might have helped in correcting for biases or confounding factors in the treatment assignment.

The ATE estimated by Propensity Score Matching increased from 0.0317 to 0.0446. The addition of the variable improved the matching process. The alternative causal inference model had the effect of improving the causal identification of the treatment effect and provided more reliable results.

Evaluating the Strengths and Limitations of the Causal Inference Model

Strengths of Our Causal Inference Model:

1. **Explicit Modeling of Causal Assumptions:** By creating a directed acyclic graph (DAG), our approach explicitly outlines the assumed relationships among variables. This is more rigorous than traditional methods, which often rely on black-box regression or ANOVA techniques that assume perfect randomization.
2. **Multiple Estimation Strategies:** The framework adopted many different methods for estimating treatment effects, such as propensity score matching and inverse probability weighting. These can help to prevent biases that can arise when randomization is less than perfect, ensuring that our estimates remain robust despite potential imbalances in the data.
3. **Sensitivity Analyses:** Our approach includes sensitivity analyses that test how robust the estimated treatment effects are against the presence of unobserved confounders. This extra layer of validation helps us understand the potential impact of hidden biases, enhancing the credibility of our causal inferences.
4. **Adaptability to Contextual Disruptions:** Causal models are designed to handle changes in context—such as new policies, interventions, or environmental shifts—that might alter the statistical properties of historical data. This adaptability makes our causal inference approach more reliable for answering causal questions in dynamic real-world settings.

Limitations and Contextual Considerations:

1. **Limited Scope of Confounders:** Our causal model, although more robust in theory, depends on the quality of the DAG and the covariates included; if key confounders are omitted, the estimates may be biased. Our DAG currently includes only a limited set of covariates (Frequency in the Past and Historical Visits), while in practice, additional factors such as Membership Status, Device Type, or Income might be required for a comprehensive model.
2. **Effectiveness of Randomization:** When randomization is perfectly executed, treatment assignment is independent of both observed and unobserved confounders, ensuring that any differences in outcomes are attributable solely to the treatment. In such ideal

circumstances, the need for extensive backdoor adjustments becomes less critical because randomization inherently balances these confounding variables across treatment groups.

3. **Oversimplification of Real-World Complexity:** Synthetic datasets are designed with predefined randomness and distributions, which means they often capture only a subset of the complexities found in real-world data. However, real-world datasets may involve intricate relationships among variables, measurement errors, missing data, and non-linear interactions that synthetic data might not fully replicate.

Comparison: Causal Inference Model vs. RCT Designs

We compared our causal inference model and RCT designs (which we used in A2) to examine the strengths and limitations of each approach in estimating treatment effects. By focusing on five main aspects—such as causal identification, estimation methods, assumption requirements, flexibility, and sensitivity to unobserved confounding—we can see how each method handles the challenges of inferring causality. This comparison helps us decide which approach is more suitable for a given context and highlights the trade-offs between model-based and design-based causal inference.

Our causal inference model is built on a clear causal diagram that shows how “Frequency in the Past” and “Historical Visits” affect both the treatment (DiscountType) and the outcome (Conversion_Rate) of this experiment. We adjust for these confounders using propensity score methods, which help to isolate the effect of the treatment. In comparison, our RCT designs mostly rely on randomization to balance observed and unobserved confounders. For example, in our Parallel Design experiment part, random assignment with blocking made sure that each group, consisting of 73 participants was comparable, reducing potential biases without the need for extensive adjustments.

In our causal model, we have average treatment effect (ATE) estimates of approximately 0.039 and 0.044 using propensity score weighting and matching. These estimates mean that the discount treatment is associated with a modest 4 percentage point increase in the conversion rate. On the other hand, as analyzed using ANCOVA in the Parallel Design, the RCT design showed a mean difference of 0.0235 with a p-value of 0.559, which is greater than 0.05, indicating that the effect was not statistically significant. However, this could also be due to various other factors like model set up, choice of variables, or handling of data.

Another critical difference between causal model and RCT designs is that the validity of our causal model depends on the assumption that all important confounders are incorporated and properly adjusted. If important confounding variables are omitted, the ATE estimate may be biased. The randomization inherent in RCT designs will be less reliant on such assumptions, as it naturally balances both observed and unobserved confounding factors. This difference also points to the importance of careful variable selection in causal models versus the inherent advantages of randomization in experimental designs.

Furthermore, the causal inference approach not only relies on choosing the appropriate confounding factors, it is also highly dependent on the accurate measurement of confounders, whereas RCT designs offer stronger internal validity through randomization. At the same time, RCT may suffer from smaller sample sizes and greater resource requirements, while a causal model can be efficient in analyzing a large sample size.

Lastly, our causal model is sensitive to unobserved confounding because its accuracy hinges on the completeness of the measured variables, whereas RCTs are less affected by such issues due to the balancing effect of random assignment.

Conclusions and Future Directions in Modern Experimental Design

Adaptive Efficiency vs. Statistical Rigor

The MAB model effectively adapted sample allocations across four discount treatments, allowing us to optimize outcomes in real time by reallocating resources to better perform treatments. While RCTs offer statistically rigorous testing and interpretability, they are less flexible and slower to adapt than MABs, which could lead to a time lag in strategy execution that reduces the effectiveness of such strategy. This contrast shows a key trade-off in experimental design, MABs prioritize performance and adaptability, while RCTs prioritize explanation and fairness.

Causal Understanding and Implications

In our causal inference model, the DAG helped illustrate the relationships between key variables such as “Frequency in the Past,” “Historical Visits,” and the assigned treatment (DiscountType). Using methods like propensity score weighting and matching, we estimated the average treatment effect (ATE) to be approximately 0.0393 and 0.0441. These results are equal to our linear regression result, where the coefficient for DiscountType was also 0.0393 and statistically significant. A sensitivity analysis was also conducted to show that unobserved confounding could shift the effect size within a range of -0.01875 to $+0.01518$, indicating both the robustness and the limitations of our model. This level of causal insight can complement the MAB model by helping explain why certain discounts performed better than others. It also emphasizes the importance of identifying and measuring confounding variables to support reliable decision-making.

Implications for A/B Testing

The results from both the MAB and causal inference models show that modern A/B testing is becoming more dynamic and insightful. The MAB model is applicable to fast-paced online settings, as it can quickly adjust based on user behavior and improve performance with zero time lag. On the other hand, causal inference models help explain why certain treatments work by

showing the causation relationships between variables. When used together, these methods allow us to run experiments that not only boost outcomes like conversion rates but also reveal what factors are actually driving the underlying user actions. This combined approach supports efficient and in-depth decision-making.

Future Steps for Research

In the future, research can combine the real-time adaptability of MABs with the statistical rigor of RCTs. This will allow data scientists to quickly optimize performance and still get reliable analysis at the same time. Adding more variables, such as user demographics, device type, or browsing history, can help improve the accuracy of results and bring the analysis more suitable to real-world users. It's also important to use sensitivity analysis to test the robustness of the findings, especially when certain influences may not be directly observable. By combining these approaches, we can build more user-centered experimental systems in research and industry settings.

Improve as a Human-Centered Data Scientist

This project has helped us grow as human-centred data scientists by showing how to balance efficiency with fairness in experimental design. Working with both Multi-Armed Bandits (MABs) and causal inference methods taught us that while MABs are great for adapting quickly and improving performance in real time, causal models are important for understanding why something works and ensuring decisions are fair and unbiased. Learning when and how to use each method will help us design smarter, more responsible experiments in the future. In today's UX research world, where data drives many decisions, having this balanced approach means we can create solutions that not only perform well but also respect users and support better overall experiences. The insights we gained from how the MAB adjusted sample allocation to how the RCT controlled for bias will guide us in making data-driven decisions that are both effective and ethical.